

Pratik Aher

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EDUCATION

State University of New York at Buffalo

Master of Engineering Science in Artificial Intelligence

December 2025

GPA: 3.2/4

Coursework: Reinforcement learning, CV, Algorithms and analysis, Data modeling and Query Language, DIC, LLM.

Savitribai Phule Pune University

Bachelor of Engineering in Artificial Intelligence and data science

May 2024

GPA: 9.2/10

Coursework: Data Structures and algorithms, Bigdata, Data Science, NLP, Pattern Recognition, ML, AI.

TECHNICAL SKILLS

Programming Languages: Python, SQL.

Deep Learning & NLP: CNNs, RNNs, LSTMs, Transformers, NLTK, PyTorch, Hugging Face, GANs, Generalized Linear Regression, TensorFlow, Scikit-learn, XGBoost, Random Forests, GenAI, LangChain, RAG, Stable Diffusion.

Data Analysis & Big Data: Pandas, NumPy, SciPy, Matplotlib, Seaborn, Apache Spark, Hadoop, Kafka, Stochastic Processes, Time Series Forecasting, Plotly, Anomaly detection, Docker, Kubernetes, Git, CI/CD, AWS(EC2/S3)

Databases & Other: MongoDB, PostgreSQL, Flask, FastAPI, MLFlow, GIT, TFX.

Certifications: Oracle AIML, AWS cloud certification, ORACLE Java Foundation and Advance, AWS Cloud Foundation, AWS Data Analytics, AWS Machine Learning Foundation.

PROFESSIONAL EXPERIENCE

GenAI Intern

Filmy AI LLC

Jersey City, New Jersey

August 2025 – Present

- **Designed and implemented an Agentic/RAG pipeline with LangChain**, integrating **vector embeddings + ChromaDB** (HNSW) retrieval and LLM generation; tuned chunking strategies and evaluated with **Recall@k/MRR**, reducing hallucinations rates by 18% and improving factual accuracy of answers by 25%.
- Fine-tuned **Gemma models via Hugging Face and PEFT/LoRA**, boosting F1 scores by **12%** and accuracy by **9%** through iterative prompt evaluation.
- **Operationalized end-to-end pipelines with FastAPI, Docker, and AWS, ensuring scalability and observability.**
- **Applied model quantization and knowledge distillation techniques within an end-to-end Generative AI system**, optimizing inference latency and scalability for real-time deployment on AWS.

Research Intern

University at Buffalo

Buffalo, New York

March 2025 – July 2025

- Integrated a real-time object detection and tracking module (YOLOv8) on the Unitree Go2 quadruped robot, achieving sub-100 ms inference latency and enabling responsive voice-command navigation through ROS2.
- Orchestrated the deployment of a ROS2-based voice control system on edge hardware, boosting object detection accuracy by 10× and ensuring reliable operation under real-world conditions.

PROJECTS

OriginGuard AI Detector

May 2025

- Built an AI based content detection system in Python to detect and quantify AI vs. human-generated content in text and images using ChatGPT Detector RoBERTa and custom paraphrase detection logic.
- Developed and deployed AI image classification to identify synthetic vs. human-captured images with **84% classification accuracy** across three benchmark datasets, using **streamlit** for real-time, user friendly content verification and improved accessibility for non technical users.

Intelligent Document Processing with Generative AI

February 2025

- Automated AI-driven document processing – Developed an end-to-end document extraction pipeline **using OCR (Tesseract, EasyOCR) and NLP (BERT, GPT-3) to classify and extract structured information (names, dates, amounts) from unstructured documents, increasing efficiency by 40% across 500+ documents weekly.**
- Deployed scalable AI-powered automation – Built and launched a FastAPI-based document processing API on AWS using Docker, enabling real-time classification and retrieval while reducing manual effort by 35%.

ACHIEVEMENTS

- **2nd Winner – AI for Good Challenge**, University at Buffalo (2025).
- Directed and mentored **Student Council & AI Student Association** organized technical, with over 10,000 participants, increasing engagement by 35% year over year.
- Managed and motivated team of 17, collaborated with 70+ companies and secured offers for 87% of students as **General Secretary of Training and Placement cell.**